Forecasting and Macroeconomic Analytics

ECON 522

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Spring 2025

Non stationarity Introduction

- There are various reasons why data may not be stationary.
- For example we saw that data with a deterministic trend cannot be weakly stationary (we also call this trend stationary).
- Data may have a break or may have a variance that change over time.
- One of the most common reasons in macro data is the presence of a stochastic trend.
- Ane example of model with a stochastic trend is a model with **unit root** also called a **random walk**.
- The reason why it is important to know if the data is stationary: Wold decomposition (thus ARMA models) and CLT

Non stationarity- Introduction

- If I knew exactly why the data is non stationary I could transform the data and make it stationary.
- For example, detrend or include breaks etc.
- We will see that this is a hard question as it is often not easy to differentiate between a stationary model (trend stationary) and a unit root (unit root with drift).
- We will see why and why this is so important.

The trend model

- As we saw many Time Series data present a trend.
- Estimation of deterministic by least squares is equivalent to estimation after linear detrending (by the FWL theorem).
- Inclusion of a linear (and possibly quadratic) time trend in a regression model is typically the easiest method to incorporate time trends.
- However, deterministic trend, while easy, is not the only possible trend: Stochastic trend has very different implications and it is hard to differentiate between the two.

Stochastic Trends

Definition 3 - Random Walk unit root $\longrightarrow$ $\boxed{\rho = 1}$

If

AR(1) $\longrightarrow$

$$\boxed{y_t = y_{t-1} + \varepsilon_t.}$$

the model is known as a random walk.

$y_t = \rho y_{t-1} + \varepsilon_t$

Definition 3 - Random Walk With Drift

If

$$y_t = c + y_{t-1} + \varepsilon_t.$$

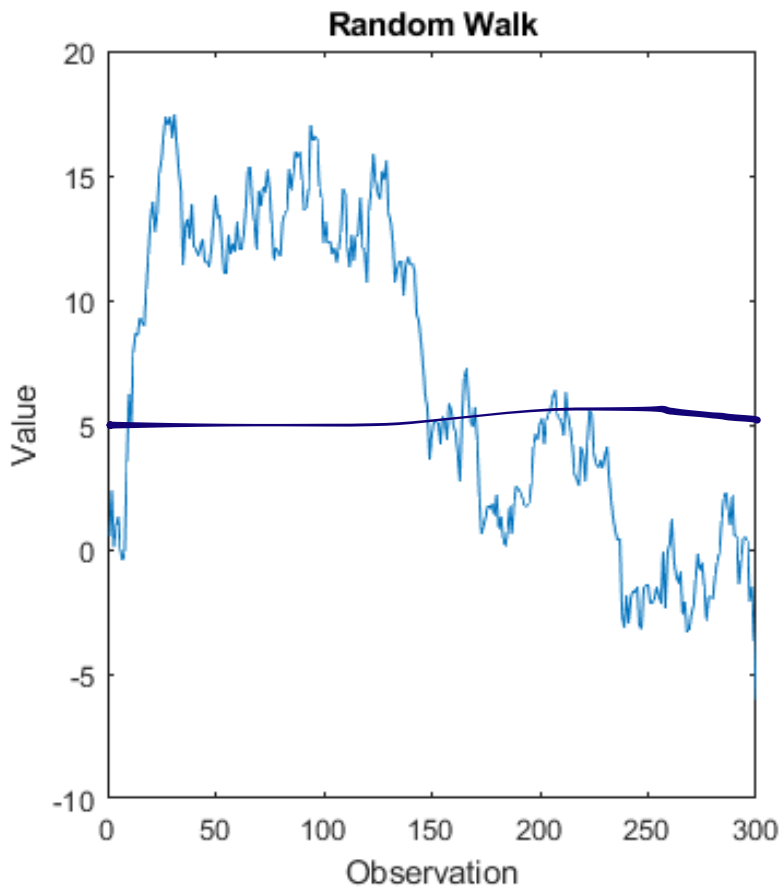
$\Delta y_t = y_t - y_{t-1} = \varepsilon_t$

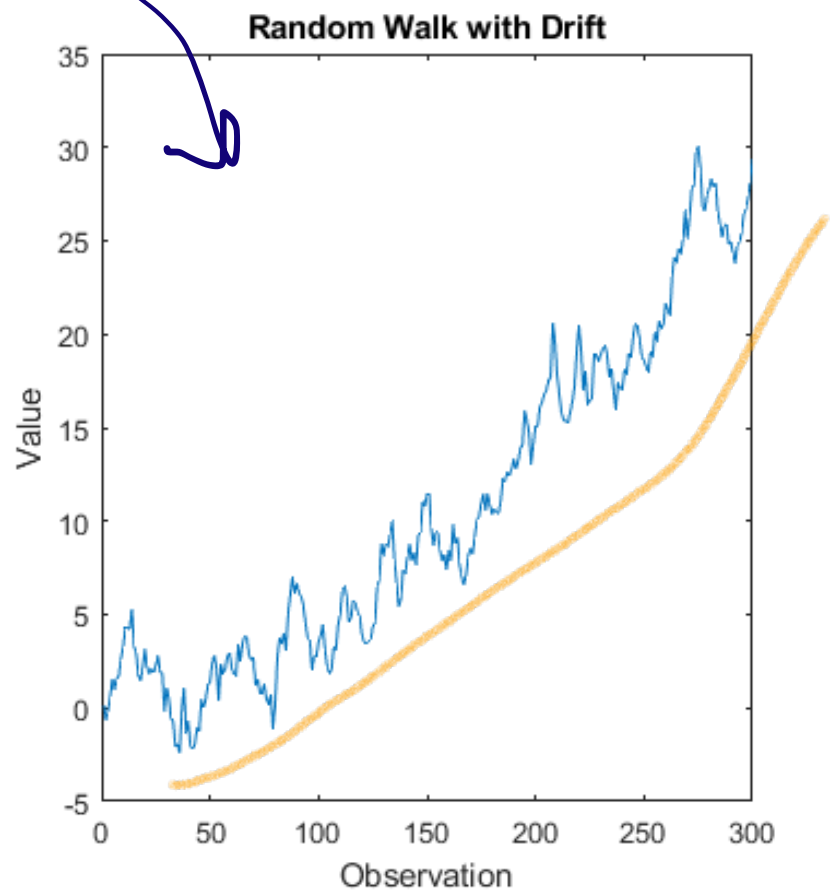
the model is known as a random walk with drift.

- Notice that in both cases the first difference would be ε_t and thus would be stationary.
- We say that y_t is integrates of order d, $I(d)$ if the dth difference is stationary.
- If the AR coefficient is > 1 the process is explosive (not a good description of most economic time series)

Stochastic Trends

deterministic or stochastic trend





Random Walk

A random Walk is also called a unit root process, a martingale, or an integrated process. By back-substitution we find

$$y_t = y_0 + \sum_{j=1}^{t} \varepsilon_j.$$

- The initial condition does not disappear for large t
- The autoregressive polynomial $\alpha(z) = 1 - z$ is not invertible, meaning that y_t cannot be written as a convergent function of the infinite past history of ε_t.

Random Walk

For Random Walk

$$y_t = y_{t-1} + \varepsilon_t = \sum_{s=1} \varepsilon_s$$

$$E\left(y_t\right) = 0, \quad \mathrm{VAR}\left(y_t\right) = t\sigma_\varepsilon^2$$

$$\gamma_k = E\left(\sum_{s=1}^{t} \varepsilon_s\right)\left(\sum_{s=1}^{t-k} \varepsilon_s\right) = \boxed{(t-k)\sigma_\varepsilon^2}$$

CON

$$\rho_k = \frac{(t-k)\sigma_\varepsilon^2}{\sqrt{t\sigma_\varepsilon^2}\sqrt{(t-k)\sigma_\varepsilon^2}} = \sqrt{\frac{t-k}{t}} \to 1$$

Random Walk with drift

For Random Walk with Drift

$$y_t = cy_{t-1} + \varepsilon_t = ct + \sum_{s=1} \varepsilon_s$$

$$E\left(y_t\right) = ct, \quad \mathrm{VAR}\left(y_t\right) = t\sigma_\varepsilon^2$$

$$\gamma_k = E\left(\sum_{s=1}^{t} \varepsilon_s\right)\left(\sum_{s=1}^{t-k} \varepsilon_s\right) = (t-k)\sigma_\varepsilon^2$$

$$\rho_k = \frac{(t-k)\sigma_\varepsilon^2}{\sqrt{t\sigma_\varepsilon^2}\sqrt{(t-k)\sigma_\varepsilon^2}} = \sqrt{\frac{t-k}{t}} \to 1$$

So they are both clearly not weakly stationary

Forecasts - Trend versus stochastic trend

Consider a trend $Y_t = \mu + ct + \Psi(L)\varepsilon_t$ versus a stochastic trend process $\Delta y_t = c + \Psi(L)\varepsilon_t$

Forecasts for Deterministic Trend

$$y_t = \alpha + ct + \varepsilon_t + \psi_1\varepsilon_{t-1} + \psi_2\varepsilon_{t-2} + \dots$$

$$y^f_{t,s} = \alpha + c(t+s) + \cancel{\psi_s\varepsilon_t + \psi_{s+1}\varepsilon_{t-1} +} \dots$$

$$\text{as} \quad s \to \infty, \quad y^f_{t,s} \to \alpha + c(t+s)$$

Forecasts for Stochastic Trend

$$\Delta y_t = c + \psi_1\varepsilon_{t-1} + \psi_2\varepsilon_{t-2} + \dots$$

$$\Delta y^f_{t,s} = c + \psi_s\varepsilon_t + \psi_{s+1}\varepsilon_{t-1} + \dots$$

$$y^f_{t,s} = sc + y_t + \cancel{(\psi_s + \psi_{s-1} + ..)\varepsilon_t + (\psi_{s+1} + \psi_s + ..)\varepsilon_{t-1}} + \dots$$

$$\text{as} \quad s \to \infty, \quad y^f_{t,s} \to sc + y_t$$

Forecasts - Trend versus stochastic trend

- For deterministic trend model the forecast converges to a linear function of the forecast horizon regardless of the value of y today.
- For the stochastic trend model the forecast is also a linear function of the forecast horizon but not the intercept changes with the starting point i.e. the best forecast for any point in the future is todat (plus a constat rate of growth equal to the drift if a RW with drift)
- These are two very different implications for how we should forecast.

MSE- Trend versus stochastic trend

It can be shown (Hamilton page 441 - do it on your own)

<u>MSE for deterministic trend</u>

$$MSE = \left(1 + \psi_1^2 + \psi_2^2 + \ldots + \psi_{s-1}^2\right)\sigma^2$$

The MSE converges to the unconditional variance of y_t.

<u>MSE for Stochastic Trend</u>

$$\begin{aligned}\text{MSE} = \Big\{1 + (1+\psi_1)^2 + (1+\psi_1+\psi_2)^2 + \ldots \\ + (1+\psi_1+\psi_2+\ldots+\psi_{s-1})^2\Big\}\end{aligned}$$

as $s \to \infty$ MSE does not converge to a fix value

Dynamic Multipliers - Trend versus stochastic trend

Dynamic Multipliers for deterministic trend

$$y_{t+s} = \mu + c(s+t) + \psi_s \varepsilon_t + \psi_{s+1}\varepsilon_{t-1} + \dots$$

$$\frac{\partial y_{t+s}}{\partial \varepsilon_t} = \psi_s \quad \text{as } s \to \infty \frac{\partial y_{t+s}}{\partial \varepsilon_t} \to 0$$

The effect of a shock vanishes eventually

Dynamic Multipliers for Stochastic Trend

$$y_{t+s} = c(s) + y_t + (\psi_s + \psi_{s-1} + ..)\,\varepsilon_t + (\psi_{s+1} + \psi_s + ..)\,\varepsilon_{t-1} + \dots$$

$$\frac{\partial y_{t+s}}{\partial \varepsilon_t} = \frac{\partial y_t}{\partial \varepsilon_t} + (\psi_s + \psi_{s-1} + \dots + \psi_1) =$$

$$= 1 + \psi_1 + \psi_2 + \dots + \psi_s$$

$$\text{as } s \to \infty \frac{\partial y_{t+s}}{\partial \varepsilon_t} \to 1 + \psi_1 + \psi_2 + \dots. = \Psi(1)$$

the effect of a shock is permanent

Transformations to Achieve Stationarity

What transformations could we do to achieve stationarity?

1. Detrend the data:

- If y_t is trend stationary then indeed $y_t - ct = \mu + \varepsilon_t$ is stationary
- If y_t is a unit root with drift, $y_t - ct = y_0 + \varepsilon_t + \varepsilon_{t-1} + \ldots + \varepsilon_1$ but this is a process with variance $t\sigma^2$: detrending removes the trend in the mean but not in the variance. Detrending is not enough.

Transformations to Achieve Stationarity

2. Take the first difference

- If y_t is a unit root with drift, $\Delta y_t = c = \varepsilon_t$. Indeed we get back to stationary process.
- If y_t is trend stationary then $\Delta y_t = c + \varepsilon_t + \varepsilon_{t-1}$ so now we have introduced a unit root in the MA part and we do not have invertibility.

Transformations to Achieve Stationarity

- There is not a single solution to transform the data to be stationary. We do really care a lot about knowing whether we have a unit root (stochastic trend) or not.
- Nelson and Plosser (1982) argued that many economic series are better characterized by a unit root than a stationary process $\rightarrow$ true or not, this shows that it is a relevant questions for empirical applications, and macro data.
- Unfortunately the issue of distinguish between these two models is not simple, especially in finite sample as they may be statistically indistinguishable or at the very least very similar.
- Testing for a unit root has become a very important step in any empirical application involving Time Series.

Unit Root Tests

Benchmark Case

$$y_t = \rho y_{t-1} + \varepsilon_t \Rightarrow \Delta y_t = \alpha y_{t-1} + \varepsilon_t \quad \alpha = (\rho - 1)$$

The hypothesis we want to test is

UR

$$H_0 : \rho = 1 \Rightarrow H_0 : \alpha = 0$$

$$H_1 : \rho < 1 \Rightarrow H_0 : \alpha < 0$$

One reasonable option would be to use the t-test (Dickey Fuller test)

$$t = \frac{\hat{\alpha}}{SE(\hat{\alpha})} = \frac{\hat{\alpha}}{\hat{\sigma}/\sqrt{\sum_{t=1}^{T} y_{t-1}^2}}$$

Unit Root Tests - Dickey Fuller Test

Unit Root Tests- Dickey Fuller test

- Under the null hypothesis, the t-test does not have the standard distribution (For a discussion on this, see next set of slides).
- Why? Random walks behave differently and have very different statistical properties. A normal is not a good approximation. (Brownian Motions)
- You can still use the t-statistics but you have to look at a different table of critical values. You cannot use the tables for the Normal. (included in most standard software).

Unit Root Tests - Dickey Fuller Test

You have two possible cases:

Intercept only

$$\Delta y_t = \mu + \rho y_{t-1} + \varepsilon_t$$

Intercept and trend

$$\Delta y_t = \mu + ct + \rho y_{t-1} + \varepsilon_t$$

The two cases have different critical values (unlike usual tests because these tests are *not invariant*) $\rightarrow$ See handout or any book.

Unit Root Tests - Dickey Fuller Test

Table of DF critical values

1. $\Delta Y_t = \mu + \alpha Y_{t-1} + u_t$ (intercept only)
2. $\Delta Y_t = \mu + ct + \alpha Y_{t-1} + u_t$ (intercept and time trend)

	Critical Values		
	10%	5%	1%
Intercept only	-2.57	-2.86	-3.43
Intercept and time trend	-3.12	-3.41	-3.96

- Reject if the $DF - t$-statistic (the t-statistic testing $\alpha = 0$) is less than the specified critical value. This is a 1 -sided test of the null hypothesis of a unit root (random walk trend) vs. the alternative that the autoregression is stationary.
- See Hamilton book for pages 762-763 for extended tables. Most softwares have packages to do these tests

Unit Root Tests - Augmented Dickey Fuller Test

- In most cases you will encounter the benchmark model we just looked at is too limited.
- There is more dynamic beside the one lag that we need to include.
- To include more dynamics you can add lags of the first difference. You can still do the same t test and look at the same table.

In an AR(p), the DF test is based on the rewritten model,

$$\Delta Y_t = \mu + \alpha Y_{t-1} + \gamma_1 \Delta Y_{t-1} + \gamma_2 \Delta Y_{t-2} + \ldots + \gamma_{p-1} \Delta Y_{t-p+1} + u_t$$

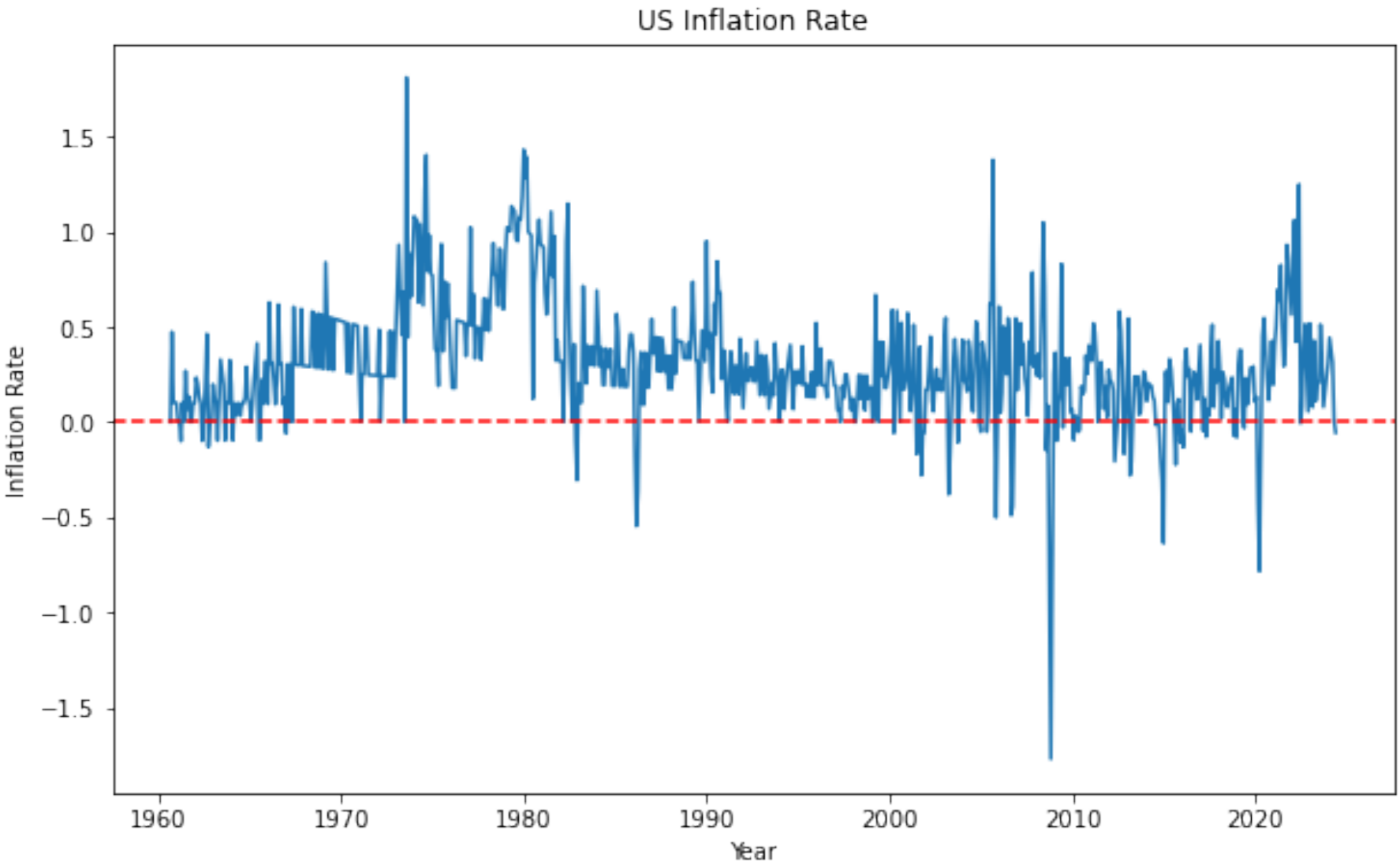
- If there is a unit root (random walk trend), $\alpha = 0$; if the AR is stationary, $\delta < 1$.
- Just estimate this regression (decide if trend or no trend) and obtain the test statistics for $\alpha = 0$.
- Reject the null hypothesis of a unit root if the t-statistic is less than the DF critical values (do not use the normal's).

Unit Root Tests - Trend or no Trend?

The decision to use the intercept-only DF test or the intercept & trend DF test depends on what the alternative is - and what the data look like.

- In the intercept-only specification, the alternative is that Y is stationary around a constant
- In the intercept and trend specification, the alternative is that Y is stationary around a linear time trend.

Example: US Inflation rate



Example: US Inflation rate

- Constant, no trend
- How many lags? (AIC, max=8)

	ADF Statistic	p-value	Number of Lags	Critical Values
1%	-3.940014	0.001759	8	-3.439018
10%	-3.940014	0.001759	8	-2.568807
5%	-3.940014	0.001759	8	-2.865366

- We reject at all values but close at 1
- These results may change if you change the sample size
- This is a topic of debate: Inflation is persistent
- you may see inflation modeled as having a unit root (more in the past than now... we will see why)

Stationarity Tests

- Why should we test the null of "non stationarity"?
- We usually pose a test with a null that we would like to not reject.
- So it may make more sense to test a null of stationarity against and alternative of non stationarity. $\rightarrow$ Stationarity tests.
- There are a few of these as well.
- NOTE: do not expect the two tests to give you the same answer necessarily. Why? We can reject for various reasons.

Summary

1. The random walk model is the workhorse model for trends in economic time series data
2. To determine whether y_t has a stochastic trend, first plot y_t, then if a trend looks plausible, test for a unit root (decide which version, intercept or intercept+trend)
3. Detecting a unit root is not easy: use a package of different tests.
4. If the test fails to reject, conclude that y_t has a unit root (random walk stochastic trend)
5. If y_t has a unit root, use Δy_t for regression analysis and forecasting.

Which test should you use?

- As we saw we can make lots of mistakes by making the wrong assumption.
- Any test will have a hard time to distinguish between a root equal to 0.95 and 1 in finite samples.
- Use the most powerful test available to you. (DF-GLS, ERS)
- You may want to present results from a battery of tests (this is what appendix is for)
- Decide if you want to test for stationarity (be prepared for conflicting results)
- The model specification matters (lags -AIC, BIC, use data dependent criteria -MAIC)
- Think about confounding reasons (structural breaks, HSK, Seasonality etc.) Looks for tests robust to these if needed.

Final Note

- This is not an easy question to ask our data so do not expect an easy answer.
- Believes have fluctuated over the years. First nobody knew, then everything has a unit root, then tests with more power let us know that most variables are persistent but not exactly unit roots (near unit root/fractional integrated).
- Make a good case and move on.
- Be careful when applying any type of filter to your data(HP, BK): Filters should not be applied to non stationary data.
- When in doubt, you are always better off NOT imposing a restrictions that may be wrong (see some of my work).
- More consensus lately on either using *robust* methods of leave things in levels especially in VARs (more on this later)
- We will see in a separate lecture why these tests do not have a standard normal asymptotic distribution.